

DBMS – PRODUCT AND CATEGORY MANAGEMENT

LENSKART – PRODUCT AND CATEGORY MANAGEMENT DATABASE SYSTEM

1. DATABASE DESIGN

1.1 CATEGORY TABLE

The Category table stores the different categories of products available in the Lenskart database.

Attribute	Description
------------------	--------------------

Category_ID	Unique category ID
-------------	--------------------

Category_Name	Name of the product category
---------------	------------------------------

SQL Query

```
CREATE TABLE Category (  
    Category_ID NUMBER PRIMARY KEY,  
    Category_Name VARCHAR2(50) NOT NULL UNIQUE  
);
```

1.2 PRODUCT TABLE

The Product table stores the details of products available in Lenskart.

Attribute	Description
------------------	--------------------

Product_ID	Unique product ID
------------	-------------------

Product_Name	Name of the product
--------------	---------------------

Brand	Product brand
-------	---------------

Price	Product price
-------	---------------

Stock	Available stock
-------	-----------------

Attribute	Description
-----------	-------------

Category_ID	Category ID associated with the product
-------------	---

```
SQL> CREATE TABLE Product (  
 2     Product_ID NUMBER PRIMARY KEY,  
 3     Product_Name VARCHAR2(100) NOT NULL,  
 4     Brand VARCHAR2(50) NOT NULL,  
 5     Price NUMBER(10,2) NOT NULL,  
 6     Stock NUMBER NOT NULL,  
 7     Category_ID NUMBER,  
 8     FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)  
 9 );  
  
Table created.
```

2. PRIMARY AND FOREIGN KEY RELATIONSHIP

The relationship between the two tables is:

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key referencing Category.Category_ID
- Relationship → **One-to-Many (1:M)**

This means one category can contain multiple products.

Relationship

CATEGORY

Category_ID (PK)

Category_Name

|

|

| 1

|

| M

↓

PRODUCT

Product_ID (PK)

Product_Name

Brand

Price

Stock

Category_ID (FK)

3. TABLE CREATION

3.1 CREATE CATEGORY TABLE

```
SQL> CREATE TABLE Category (  
2     Category_ID NUMBER PRIMARY KEY,  
3     Category_Name VARCHAR2(50) NOT NULL UNIQUE  
4 );  
  
Table created.
```

3.2 CREATE PRODUCT TABLE

```
SQL> CREATE TABLE Product (  
2     Product_ID NUMBER PRIMARY KEY,  
3     Product_Name VARCHAR2(100) NOT NULL,  
4     Brand VARCHAR2(50) NOT NULL,  
5     Price NUMBER(10,2) NOT NULL,  
6     Stock NUMBER NOT NULL,  
7     Category_ID NUMBER,  
8     FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)  
9 );
```

4. DATA INSERTION

4.1 CATEGORY

RECORDS

```
SQL> INSERT INTO Category (Category_ID, Category_Name)
  2 VALUES (1, 'Eyeglasses');

1 row created.

SQL>
SQL> INSERT INTO Category (Category_ID, Category_Name)
  2 VALUES (2, 'Sunglasses');

1 row created.

SQL>
SQL> INSERT INTO Category (Category_ID, Category_Name)
  2 VALUES (3, 'Contact Lenses');

1 row created.

SQL>
SQL> INSERT INTO Category (Category_ID, Category_Name)
  2 VALUES (4, 'Kids Eyewear');

1 row created.
```

4.2 PRODUCT RECORDS

```
SQL> INSERT INTO Product
  2 (Product_ID, Product_Name, Brand, Price, Stock, Category_ID)
  3 VALUES
  4 (101, 'Air Flex Eyeglasses', 'Lenskart Air', 1500.00, 25, 1);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 (Product_ID, Product_Name, Brand, Price, Stock, Category_ID)
  3 VALUES
  4 (102, 'Classic Round Eyeglasses', 'Vincent Chase', 2000.00, 15, 1);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 (Product_ID, Product_Name, Brand, Price, Stock, Category_ID)
  3 VALUES
  4 (103, 'Polarized Sunglasses', 'Vincent Chase', 1800.00, 20, 2);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 (Product_ID, Product_Name, Brand, Price, Stock, Category_ID)
  3 VALUES
  4 (104, 'Daily Contact Lenses', 'Aqualens', 900.00, 30, 3);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 (Product_ID, Product_Name, Brand, Price, Stock, Category_ID)
  3 VALUES
  4 (105, 'Kids Round Eyeglasses', 'Lenskart', 1200.00, 10, 4);

1 row created.
```

5. DATA RETRIEVAL

5.1 DISPLAY CATEGORY RECORDS

```
SQL> SELECT * FROM Category;
```

```
CATEGORY_ID CATEGORY_NAME
```

```
-----  
1 Eyeglasses  
2 Sunglasses  
3 Contact Lenses  
4 Kids Eyewear
```

5.2 DISPLAY PRODUCT RECORDS

```
SQL> SELECT * FROM Product;
```

```
PRODUCT_ID
```

```
-----  
PRODUCT_NAME
```

```
-----  
BRAND
```

```
PRICE
```

```
STOCK
```

```
-----  
CATEGORY_ID
```

```
-----  
101  
Air Flex Eyeglasses  
Lenskart Air  
1
```

```
1500
```

```
25
```

```
PRODUCT_ID
```

```
-----  
PRODUCT_NAME
```

```
-----  
BRAND
```

```
PRICE
```

```
STOCK
```

```
-----  
CATEGORY_ID
```

```
-----  
102  
Classic Round Eyeglasses  
Vincent Chase  
1
```

```
2000
```

```
15
```

```
PRODUCT_ID
```

```
-----  
PRODUCT_NAME
```

```
-----  
BRAND
```

```
PRICE
```

```
STOCK
```

```
-----  
CATEGORY_ID
```

```
-----  
103  
Polarized Sunglasses  
Vincent Chase  
2
```

```
1800
```

```
20
```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND                                     PRICE      STOCK
-----
CATEGORY_ID
-----
      104
Daily Contact Lenses
Aqualens                                     900         30
      3

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND                                     PRICE      STOCK
-----
CATEGORY_ID
-----
      105
Kids Round Eyeglasses
Lenskart                                     1200        10
      4

```

6. PRODUCT UPDATE

```

SQL>
SQL> UPDATE Product
  2  SET Price = 1600.00,
  3      Stock = 30
  4  WHERE Product_ID = 101;

1 row updated.

```

To verify the updated product:

```

SQL> SELECT * FROM Product
  2  WHERE Product_ID = 101;

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND                                     PRICE      STOCK
-----
CATEGORY_ID
-----
      101
Air Flex Eyeglasses
Lenskart Air                                     1600        30
      1

```

7. PRODUCT DELETION

```

SQL>
SQL> DELETE FROM Product
  2  WHERE Product_ID = 105;

1 row deleted.

```

To verify the deletion:

```

SQL>
SQL> SELECT * FROM Product;

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND                                     PRICE      STOCK
-----
CATEGORY_ID
-----
      101
Air Flex Eyeglasses
Lenskart Air                               1600        30
      1

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND                                     PRICE      STOCK
-----
CATEGORY_ID
-----
      102
Classic Round Eyeglasses
Vincent Chase                             2000        15
      1

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND                                     PRICE      STOCK
-----
CATEGORY_ID
-----
      103
Polarized Sunglasses
Vincent Chase                             1800        20
      2

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND                                     PRICE      STOCK
-----
CATEGORY_ID
-----
      104
Daily Contact Lenses
Aqualens                                     900        30
      3

```

```

SQL> UPDATE Category
  2  SET Category_Name = 'Kids Glasses'
  3  WHERE Category_ID = 4;

1 row updated.

```

To verify the update:

```

SQL> SELECT * FROM Category
  2  WHERE Category_ID = 4;

CATEGORY_ID CATEGORY_NAME
-----
      4 Kids Glasses

```

9. CATEGORY-WISE PRODUCT REPORT

```

SQL>
SQL> SELECT
  2      c.Category_Name,
  3      p.Product_ID,
  4      p.Product_Name,
  5      p.Brand,
  6      p.Price,
  7      p.Stock
  8  FROM Category c
  9  JOIN Product p
 10  ON c.Category_ID = p.Category_ID
 11  ORDER BY c.Category_Name;

```

Output

```

CATEGORY_NAME                                     PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND                                     PRICE      STOCK
-----
Contact Lenses                                     104
Daily Contact Lenses
Aqualens                                     900        30

Eyeglasses                                     101
Air Flex Eyeglasses
Lenskart Air                                     1600        30

```

CATEGORY_NAME	PRODUCT_ID
PRODUCT_NAME	
BRAND	PRICE STOCK
Eyeglasses	102
Classic Round Eyeglasses	
Vincent Chase	2000 15
Sunglasses	103
Polarized Sunglasses	
CATEGORY_NAME	PRODUCT_ID
PRODUCT_NAME	
BRAND	PRICE STOCK
Vincent Chase	1800 20

10. CATEGORY-WISE TOTAL PRODUCTS

```

SQL> SELECT
2      c.Category_Name,
3      COUNT(p.Product_ID) AS Total_Products
4  FROM Category c
5  LEFT JOIN Product p
6  ON c.Category_ID = p.Category_ID
7  GROUP BY c.Category_Name
8  ORDER BY c.Category_Name;

```

REPORT

Output

CATEGORY_NAME	TOTAL_PRODUCTS
Contact Lenses	1
Eyeglasses	2
Kids Glasses	0
Sunglasses	1